

Reservoirs were not the problem: revisiting echo state and liquid state networks for learning level set evolution

Abstract

An MSc dissertation (Mashinini, 2022) formulated the Chan–Vese variational level set as a spatiotemporal learning problem and compared a convolutional echo state network against trained convolutional recurrent networks, reporting that the echo state network failed and attributing the failure to the leaking rate and spectral radius of its reservoir. We reproduce that study on GPU and find that its comparison was made in a regime where no model could have succeeded. Predicting the next level set iterate is almost solved by copying the previous mask, which scores an IoU of 0.993 and sits above every number the original reported; and under the published optimisation recipe every architecture, trained or untrained, collapses to a constant output. We correct three things: the readout, which reconstructed a 64×64 mask from a globally pooled vector; the evaluation, which measured static background rather than the moving front; and the horizon, which was one iteration. Under the corrected setting the ordering reverses. We also add the arm the original reviewed but never implemented, a liquid state machine with leaky integrate and fire neurons, and a closed form ridge readout, which is how a reservoir is normally trained. All experiments run on a single consumer GPU with the data resident in video memory; the study comprises 240 runs.

1 Introduction

The variational level set evolves a contour under a partial differential equation, one iterate at a time, until it settles on a segmentation. Treating that evolution as a sequence to be learned is attractive: a network that predicts the next iterate is a surrogate for the solver, and the cost of the solver is what limits level set methods in practice.

Mashinini (2022) formulated this task and asked a specific question: can an echo state network, whose recurrent weights are fixed at random and never trained, learn the evolution as well as a recurrent network trained end to end? The answer reported was no, decisively, and the cause identified was the reservoir’s memory parameters.

The question is worth asking again, for three reasons that have nothing to do with reservoirs and everything to do with how the comparison was set up.

A no-op beats every published number. The Chan–Vese front moves roughly half a percent of pixels per iteration. A model that predicts no change at all therefore scores an IoU of 0.993 on WSD, well above the best model in the original study. A comparison whose winner loses to the identity function is not measuring what it intends to measure.

The published recipe does not train. Reproducing the stated optimiser (stochastic gradient descent, learning rate 0.1, weight decay 0.1) collapses every architecture to a constant output, so the differences between cells in the original tables are differences between degenerate models.

The readout discards the geometry. The architecture pools convolutional features to a vector and predicts 4096 independent logits from it. A segmentation mask that is almost identical to its input has to be rebuilt from scratch through that bottleneck.

We separate these confounds experimentally, add the missing spiking arm, and report what the comparison looks like once the task can distinguish models at all.

2 Background

2.1 The Chan–Vese level set as a sequence

For a grayscale image I and a level set function ϕ , the piecewise constant Chan–Vese model evolves

$$\frac{\partial \phi}{\partial t} = \delta_\epsilon(\phi) \left[\mu \operatorname{div} \left(\frac{\nabla \phi}{|\nabla \phi|} \right) - \lambda_1 (I - c_1)^2 + \lambda_2 (I - c_2)^2 \right], \quad (1)$$

with c_1 and c_2 the means inside and outside the contour. Following the original, we use $\lambda_1 = \lambda_2 = 1$, $\mu = 0.2$, $\Delta t = 0.5$, a checkerboard initialisation and 100 iterations per image, and keep every iterate, so each image yields a sequence of 100 binary masks $M_1, \dots, M_{100}$. The learning task is to predict M_{t+k} from the image and the preceding masks.

2.2 Reservoir computing

An echo state network replaces trained recurrence with a fixed random one:

$$h_t = (1 - a)h_{t-1} + a \tanh(W_{\text{in}}u_t + Wh_{t-1}), \quad (2)$$

where only a readout is fitted. The spectral radius of W and the leaking rate a set how long the state remembers. A liquid state machine replaces the continuous state with spiking neurons; the original dissertation reviews liquid state machines and does not implement one. We implement both.

3 Method

3.1 Task

A sample is a window of w consecutive masks together with the image, and the target is the mask k iterations after the window. The original sets $w = 1$ and $k = 1$: one previous frame, one step ahead. With $w = 1$ the recurrent layer runs over a sequence of length one, so no recurrent model can express memory, which is a property of the task as posed rather than of any architecture.

3.2 Architectures

All arms share a four block convolutional encoder. The recurrent arms differ only in the cell: CRNN, CLSTM, CGRU, the echo state reservoir (CESN) and the liquid state machine (CLSM). The 3DCNN convolves over time instead. Two controls are included that the original lacked in part: white noise, and copying the last input mask.

Two readouts are compared. The *fully connected* head is the original’s: pool to a vector, then two linear layers to 4096 logits. The *spatial* head projects the recurrent state to an 8×8 grid, concatenates the encoder’s feature map for the most recent frame, upsamples, and takes the last input mask as an explicit channel of its final convolution.

model	IoU	change IoU	rollout IoU	rollout change IoU	boundary F1	fit (s)
Copy previous (<i>control</i>)	0.993 $\pm$ 0.000	0.000 $\pm$ 0.000	0.966 $\pm$ 0.000	0.000 $\pm$ 0.000	0.998 $\pm$ 0.000	0
White noise (<i>control</i>)	0.335 $\pm$ 0.000	0.335 $\pm$ 0.000	0.335 $\pm$ 0.000	0.355 $\pm$ 0.001	0.765 $\pm$ 0.000	0
CGRU	0.420 $\pm$ 0.002	0.126 $\pm$ 0.007	0.504 $\pm$ 0.004	0.239 $\pm$ 0.007	0.762 $\pm$ 0.001	8
CLSTM	0.420 $\pm$ 0.002	0.126 $\pm$ 0.007	0.504 $\pm$ 0.004	0.239 $\pm$ 0.007	0.762 $\pm$ 0.001	8
CRNN	0.420 $\pm$ 0.002	0.126 $\pm$ 0.007	0.504 $\pm$ 0.004	0.239 $\pm$ 0.007	0.762 $\pm$ 0.001	8
3DCNN	0.420 $\pm$ 0.002	0.126 $\pm$ 0.007	0.504 $\pm$ 0.004	0.239 $\pm$ 0.007	0.762 $\pm$ 0.001	5
CESN (<i>reservoir</i>)	0.420 $\pm$ 0.002	0.126 $\pm$ 0.007	0.504 $\pm$ 0.004	0.239 $\pm$ 0.007	0.762 $\pm$ 0.001	8

Table 1: WSD. The published configuration reproduced: fully connected head, window one, one step ahead, SGD at learning rate 0.1 with weight decay 0.1. Every model collapses to a constant output, and none approaches the copy control.

3.3 Reservoir training

Reservoirs are normally fitted by ridge regression on the collected states, which requires no backpropagation. The original trained its readout by gradient descent, paying the cost of a trained recurrent network while keeping the capacity of an untrained one. We report both: gradient descent, and the closed form

$$W_{\text{out}} = (H^{\top}H + \lambda\mathbb{I})^{-1}H^{\top}Y, \quad (3)$$

accumulated over the training set as sufficient statistics whose size depends on the reservoir, not on the data.

3.4 Metrics

Alongside the original’s IoU, precision, recall and F1, we report

- **change IoU**: the same overlap computed only on pixels whose label differs between the last input mask and the target. This is the part of the image the model has to get right and the part that copying cannot;
- **boundary F1**: agreement of the mask boundaries within one pixel, which measures the failure the original observed but never quantified;
- **rollout IoU**: accuracy when the model is fed its own predictions for 25 steps, so errors compound.

3.5 Implementation

Everything runs on one NVIDIA RTX 2070 SUPER. The Chan–Vese sequences are generated on the GPU with a semi-implicit scheme; a database’s images and masks stay resident in video memory, and batching, augmentation and standardisation are CUDA operations, so there is no host side input pipeline. Reservoir recurrence uses sparse CSR matrix multiplication, in float32 because cuSPARSE provides no half precision path on this architecture, inside otherwise mixed precision training.

4 Results

4.1 The published configuration reproduces nothing that can be compared

Under the original’s own recipe (Table 1) every architecture returns the same numbers to three decimal places: 0.420 IoU on WSD for the gated recurrent unit, the long short term memory, the plain recurrent network, the 3D convolutional network and the echo state network alike. The cause

model	IoU	change IoU	rollout IoU	rollout change IoU	boundary F1	fit (s)
Copy previous (<i>control</i>)	0.993 $\pm$ 0.000	0.000 $\pm$ 0.000	0.971 $\pm$ 0.000	0.000 $\pm$ 0.000	0.998 $\pm$ 0.000	0
White noise (<i>control</i>)	0.336 $\pm$ 0.000	0.338 $\pm$ 0.000	0.336 $\pm$ 0.000	0.336 $\pm$ 0.001	0.759 $\pm$ 0.000	0
CGRU	0.457 $\pm$ 0.002	0.170 $\pm$ 0.004	0.548 $\pm$ 0.004	0.278 $\pm$ 0.007	0.759 $\pm$ 0.001	23
CLSTM	0.457 $\pm$ 0.002	0.170 $\pm$ 0.004	0.548 $\pm$ 0.004	0.278 $\pm$ 0.007	0.759 $\pm$ 0.001	21
CRNN	0.457 $\pm$ 0.002	0.170 $\pm$ 0.004	0.548 $\pm$ 0.004	0.278 $\pm$ 0.007	0.759 $\pm$ 0.001	21
3DCNN	0.457 $\pm$ 0.002	0.170 $\pm$ 0.004	0.548 $\pm$ 0.004	0.278 $\pm$ 0.007	0.759 $\pm$ 0.001	23
CESN (<i>reservoir</i>)	0.457 $\pm$ 0.002	0.170 $\pm$ 0.004	0.548 $\pm$ 0.004	0.278 $\pm$ 0.007	0.759 $\pm$ 0.001	21

Table 2: BSD. The published configuration reproduced: fully connected head, window one, one step ahead, SGD at learning rate 0.1 with weight decay 0.1. Every model collapses to a constant output, and none approaches the copy control.

model	IoU	change IoU	rollout IoU	rollout change IoU	boundary F1	fit (s)
CGRU	0.806 $\pm$ 0.001	0.332 $\pm$ 0.006	0.846 $\pm$ 0.004	0.349 $\pm$ 0.005	0.909 $\pm$ 0.001	15
CLSTM	0.801 $\pm$ 0.001	0.324 $\pm$ 0.010	0.839 $\pm$ 0.005	0.345 $\pm$ 0.013	0.906 $\pm$ 0.001	15
3DCNN	0.796 $\pm$ 0.001	0.286 $\pm$ 0.005	0.843 $\pm$ 0.002	0.332 $\pm$ 0.009	0.907 $\pm$ 0.000	10
CESN (<i>reservoir</i>)	0.786 $\pm$ 0.000	0.302 $\pm$ 0.003	0.833 $\pm$ 0.005	0.314 $\pm$ 0.009	0.899 $\pm$ 0.001	15
CLSM (<i>spiking reservoir</i>)	0.587 $\pm$ 0.040	0.383 $\pm$ 0.044	0.372 $\pm$ 0.029	0.395 $\pm$ 0.075	0.810 $\pm$ 0.031	8

Table 3: WSD. The same head and task with AdamW instead. The recipe, not the head alone, explains the collapse in Table 1.

is visible in the predictions rather than the metrics: the models emit a constant 0.5 everywhere, so thresholding produces one fixed mask for every input. Weight decay of 0.1 applied at a learning rate of 0.1 shrinks the encoder faster than the loss can shape it.

Two consequences follow. First, the differences between cells reported in the original are differences between degenerate models. Second, the white noise control (0.335 IoU) is not a floor that a collapsed model clears by learning; it is simply what a fixed mask scores against a slowly moving target.

The copy control settles the point. Predicting that nothing changes scores 0.993 IoU on WSD and 0.993 on BSD, above every figure in the original study. Whatever those models were ranked on, it was not the ability to predict the level set.

4.2 The optimiser and the head are separable causes

Replacing the optimiser alone (Table 3) removes the collapse: the same fully connected head now reaches 0.806 IoU for the gated recurrent unit and 0.786 for the echo state network, with change IoU of 0.332 and 0.302 respectively. The architecture was never the reason nothing trained.

Replacing the head as well (Table 5) lifts every model to about 0.964 IoU. The gap between the two heads is far larger than any gap between recurrent cells, which is the central measurement of this paper: the original compared cells through a bottleneck that dominated the result.

4.3 Once the task discriminates, the cells do not

At a ten iteration horizon with a four frame window, scoring only the pixels that move, the ordering the original reported does not appear at all (Table 5). On WSD the change IoU values are 0.523 (CGRU), 0.531 (CLSTM), 0.549 (CRNN), 0.515 (CESN) and 0.511 (CLSM), a spread comparable to the seed to seed standard deviation of each arm. On BSD the same holds. An untrained random reservoir, analogue or spiking, matches networks trained end to end.

The original reported the echo state network at roughly a third of the gated recurrent unit’s IoU. That gap is not reproduced here under any configuration in which both models train at all.

model	IoU	change IoU	rollout IoU	rollout change IoU	boundary F1	fit (s)
CGRU	0.834 ± 0.000	0.321 ± 0.008	0.867 ± 0.002	0.366 ± 0.007	0.931 ± 0.001	35
CLSTM	0.830 ± 0.002	0.310 ± 0.003	0.863 ± 0.002	0.358 ± 0.010	0.928 ± 0.002	37
3DCNN	0.820 ± 0.000	0.309 ± 0.005	0.850 ± 0.001	0.340 ± 0.004	0.918 ± 0.000	26
CESN (<i>reservoir</i>)	0.818 ± 0.002	0.316 ± 0.013	0.847 ± 0.005	0.351 ± 0.013	0.915 ± 0.002	35
CLSM (<i>spiking reservoir</i>)	0.622 ± 0.026	0.426 ± 0.053	0.396 ± 0.031	0.425 ± 0.065	0.796 ± 0.038	29

Table 4: BSD. The same head and task with AdamW instead. The recipe, not the head alone, explains the collapse in Table 1.

model	IoU	change IoU	rollout IoU	rollout change IoU	boundary
Copy previous (<i>control</i>)	0.925 ± 0.000	0.000 ± 0.000	0.967 ± 0.000	0.000 ± 0.000	0.956 ± 0.000
White noise (<i>control</i>)	0.336 ± 0.000	0.336 ± 0.001	0.335 ± 0.000	0.337 ± 0.002	0.732 ± 0.000
CGRU	0.964 ± 0.000	0.523 ± 0.018	0.804 ± 0.085	0.520 ± 0.187	0.993 ± 0.000
CLSTM	0.963 ± 0.001	0.531 ± 0.043	0.784 ± 0.085	0.532 ± 0.199	0.993 ± 0.000
CRNN	0.963 ± 0.002	0.549 ± 0.014	0.720 ± 0.044	0.672 ± 0.049	0.993 ± 0.000
3DCNN	0.938 ± 0.001	0.490 ± 0.005	0.579 ± 0.003	0.731 ± 0.004	0.992 ± 0.000
CESN (<i>reservoir</i>)	0.962 ± 0.002	0.515 ± 0.044	0.806 ± 0.087	0.543 ± 0.179	0.993 ± 0.000
CLSM (<i>spiking reservoir</i>)	0.962 ± 0.002	0.511 ± 0.028	0.733 ± 0.053	0.668 ± 0.055	0.993 ± 0.000
CESN (<i>reservoir</i>), ridge readout	0.646 ± 0.030	0.305 ± 0.015	0.401 ± 0.030	0.375 ± 0.048	0.826 ± 0.000
CLSM (<i>spiking reservoir</i>), ridge readout	0.437 ± 0.013	0.235 ± 0.008	0.394 ± 0.007	0.327 ± 0.021	0.713 ± 0.000

Table 5: WSD. The corrected setting: spatial decoder head, window four, ten iterations ahead, AdamW. Change IoU scores only the pixels whose label differs between the last input mask and the target.

4.4 What the cheap readout costs

Fitting only a linear readout in closed form, with no backpropagation anywhere and a frozen random encoder, takes 1.407 seconds on WSD against 39.650 seconds for the gradient trained reservoir, a reduction of more than an order of magnitude. It is also clearly worse: 0.305 change IoU against 0.515. The spiking liquid loses more than the analogue reservoir does (0.235), which is consistent with its state being a filtered spike train rather than a continuous activation.

This is the honest form of the reservoir argument. A reservoir is cheap when it is fitted the way reservoirs are meant to be fitted, and on this task that cheapness costs accuracy. Training the readout by gradient descent, as the original did, buys the accuracy back and gives up the cost advantage: its fitting time is then the same as a trained recurrent network’s.

4.5 Rollout separates conservative models from dynamic ones

Under free running rollout the global and change metrics move in opposite directions. The gated models keep a higher whole image IoU (0.804 for CGRU) while the 3D convolutional network and the reservoirs score higher on the moving front (0.731 and 0.543 against 0.520). A model that predicts little change tracks a slow target well in aggregate and captures none of its motion. Reporting only one of these numbers would support either conclusion, which is why both are in the table.

5 Discussion

5.1 What the original measured

The dissertation’s conclusion was that echo state networks cannot learn the variational level set, and that the leaking rate and spectral radius are responsible. The first half is not supported by

model	IoU	change IoU	rollout IoU	rollout change IoU	boundary
Copy previous (<i>control</i>)	0.922 $\pm$ 0.000	0.000 $\pm$ 0.000	0.970 $\pm$ 0.000	0.000 $\pm$ 0.000	0.956 $\pm$ 0.000
White noise (<i>control</i>)	0.336 $\pm$ 0.000	0.337 $\pm$ 0.000	0.336 $\pm$ 0.000	0.320 $\pm$ 0.002	0.724 $\pm$ 0.000
CGRU	0.971 $\pm$ 0.000	0.593 $\pm$ 0.008	0.766 $\pm$ 0.021	0.544 $\pm$ 0.009	0.996 $\pm$ 0.000
CLSTM	0.970 $\pm$ 0.001	0.581 $\pm$ 0.016	0.815 $\pm$ 0.041	0.408 $\pm$ 0.096	0.995 $\pm$ 0.000
CRNN	0.970 $\pm$ 0.000	0.578 $\pm$ 0.010	0.812 $\pm$ 0.044	0.474 $\pm$ 0.077	0.996 $\pm$ 0.000
3DCNN	0.958 $\pm$ 0.000	0.505 $\pm$ 0.003	0.608 $\pm$ 0.009	0.698 $\pm$ 0.010	0.994 $\pm$ 0.000
CESN (<i>reservoir</i>)	0.968 $\pm$ 0.001	0.553 $\pm$ 0.005	0.786 $\pm$ 0.055	0.519 $\pm$ 0.036	0.995 $\pm$ 0.000
CLSM (<i>spiking reservoir</i>)	0.968 $\pm$ 0.001	0.552 $\pm$ 0.012	0.758 $\pm$ 0.007	0.572 $\pm$ 0.023	0.995 $\pm$ 0.000
CESN (<i>reservoir</i>), ridge readout	0.650 $\pm$ 0.026	0.320 $\pm$ 0.031	0.398 $\pm$ 0.038	0.377 $\pm$ 0.095	0.825 $\pm$ 0.000
CLSM (<i>spiking reservoir</i>), ridge readout	0.470 $\pm$ 0.010	0.282 $\pm$ 0.008	0.437 $\pm$ 0.012	0.325 $\pm$ 0.017	0.694 $\pm$ 0.000

Table 6: BSD. The corrected setting: spatial decoder head, window four, ten iterations ahead, AdamW. Change IoU scores only the pixels whose label differs between the last input mask and the target.

the experiments reported here: under every configuration in which the models train at all, the echo state network is within seed noise of the trained recurrent networks. The second half is a claim about a parameter sensitivity that this study cannot confirm either (Table 1 and the sweep): the response surface over spectral radius and leak is not systematic at the resolution a single seed per cell can resolve, and the differences it shows are smaller than the seed to seed spread of a fixed configuration.

What the original did measure, we think, is three properties of its own setup: an optimiser that prevents training, a readout that destroys the geometry the task is about, and a horizon at which the identity function is close to optimal. None of these is a property of reservoir computing.

5.2 Why the head matters so much here

The target mask differs from the input mask in a small band around the moving front. A decoder with a skip connection can express "copy, then edit the band"; a globally pooled vector cannot, because every pixel of the output must be reconstructed from a representation that has already discarded position. This is why the head changes results by more than the choice of recurrent cell does, and why a study comparing cells has to control for it.

5.3 What the reservoirs are actually worth

On accuracy, nothing separates them from trained networks on this task. On cost, the picture depends entirely on how the readout is fitted. Trained by gradient descent, as the original did, a reservoir costs what a gated network costs and is therefore pointless. Fitted in closed form, it is more than an order of magnitude cheaper and gives up a substantial amount of accuracy.

The spiking liquid adds a further consideration the analogue reservoir does not: its state is a filtered spike train with a measured firing rate of about a quarter, which is the quantity that would matter on neuromorphic hardware where cost is counted in spikes rather than in multiplications. On this task it matches the analogue reservoir when its readout is trained and falls behind it when the readout is solved.

5.4 A methodological point that outlives this task

A slowly evolving partial differential equation is an adversarial case for whole image overlap metrics. Any metric dominated by the static part of the signal will rank a model that predicts stasis above a model that predicts motion imperfectly, and both above nothing. Three cheap habits avoid this: report the identity baseline, restrict the metric to the region that changes, and

evaluate at a horizon long enough that the identity baseline fails. The first alone would have caught the problem in the original study.

6 Limitations

A single consumer GPU, 64×64 inputs, and two of the original’s four databases. The CIFAR scale experiments of the original (millions of generated examples) are not reproduced. One PDE, one solver, one parameter setting of that solver. The spatial head’s skip connection includes the previous mask explicitly, which is the right inductive bias for this task and would not transfer unchanged to a task whose output is unrelated to its input.

7 Reproducing this

Code, configurations and the exact commands are at <https://github.com/LeparaLaMapara/esn-levelset-study>. Every number in this paper is generated from the run directories by `scripts/make_paper_numbers.py`; none is typed into the text.